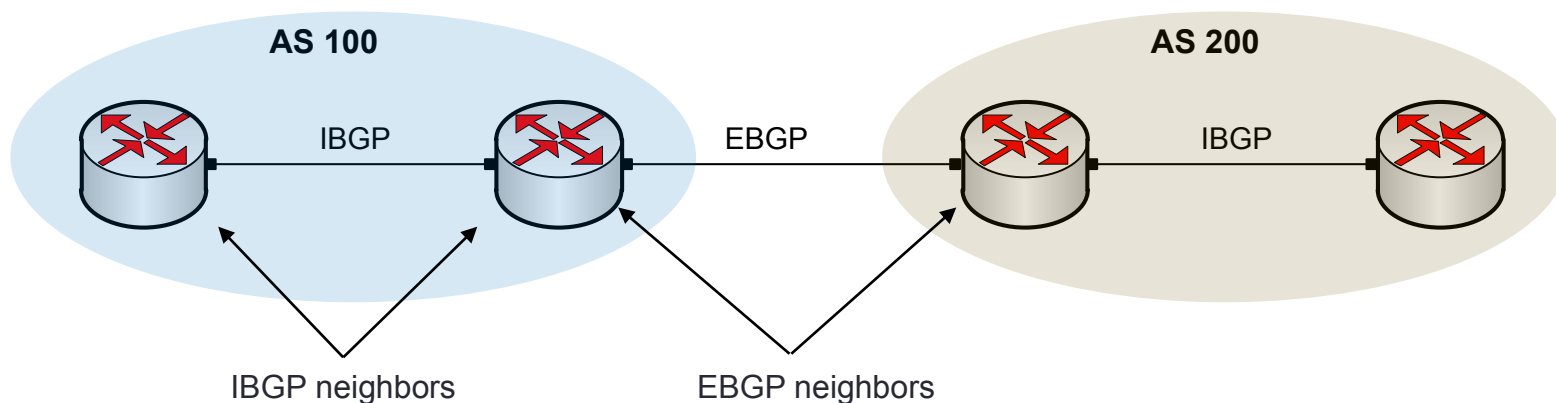


What will be covered ?

- Attacks on BGP protocol and security mechanisms
- Class 1: BGP basics
- Class 1: BGP Authentication
- Class 2: BGP Hijacking and mitigation techniques
- Class 2: IP spoofing, DoS attacks, and mitigation techniques

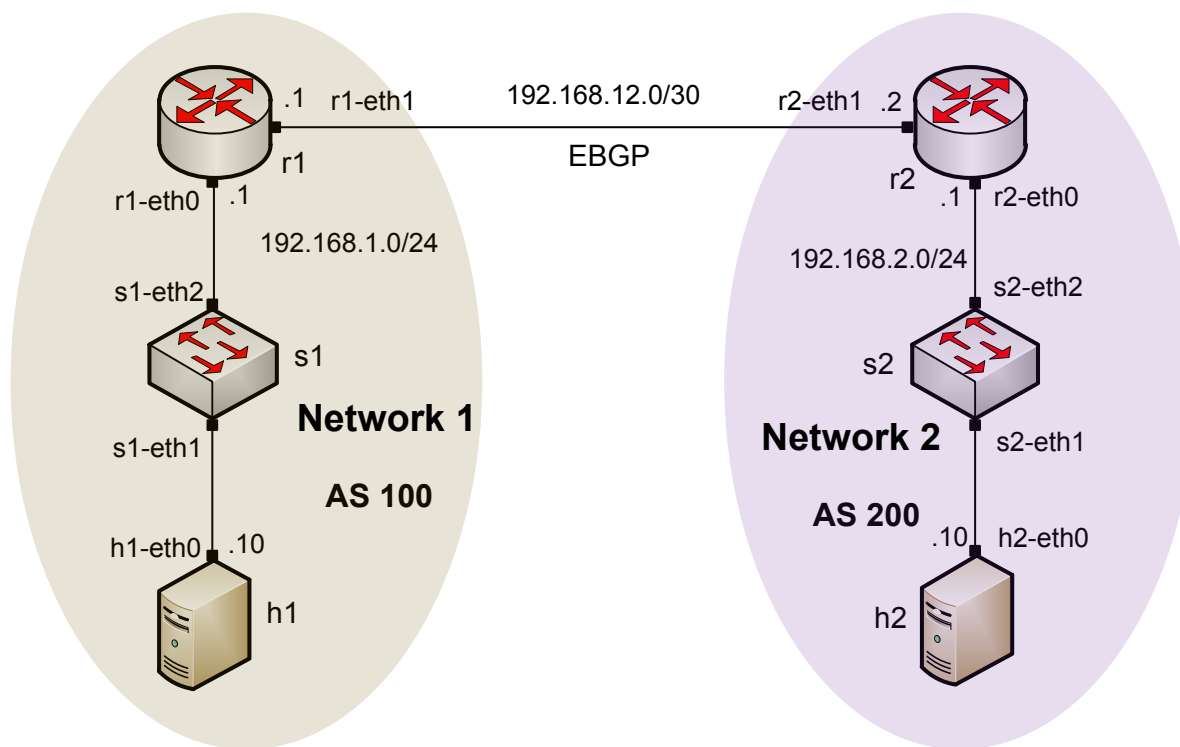
What is BGP?

- The Internet can be viewed as a collection of networks or Autonomous Systems (ASes) that are interconnected
- BGP is an exterior gateway protocol designed to exchange routing and reachability information among ASes on the Internet



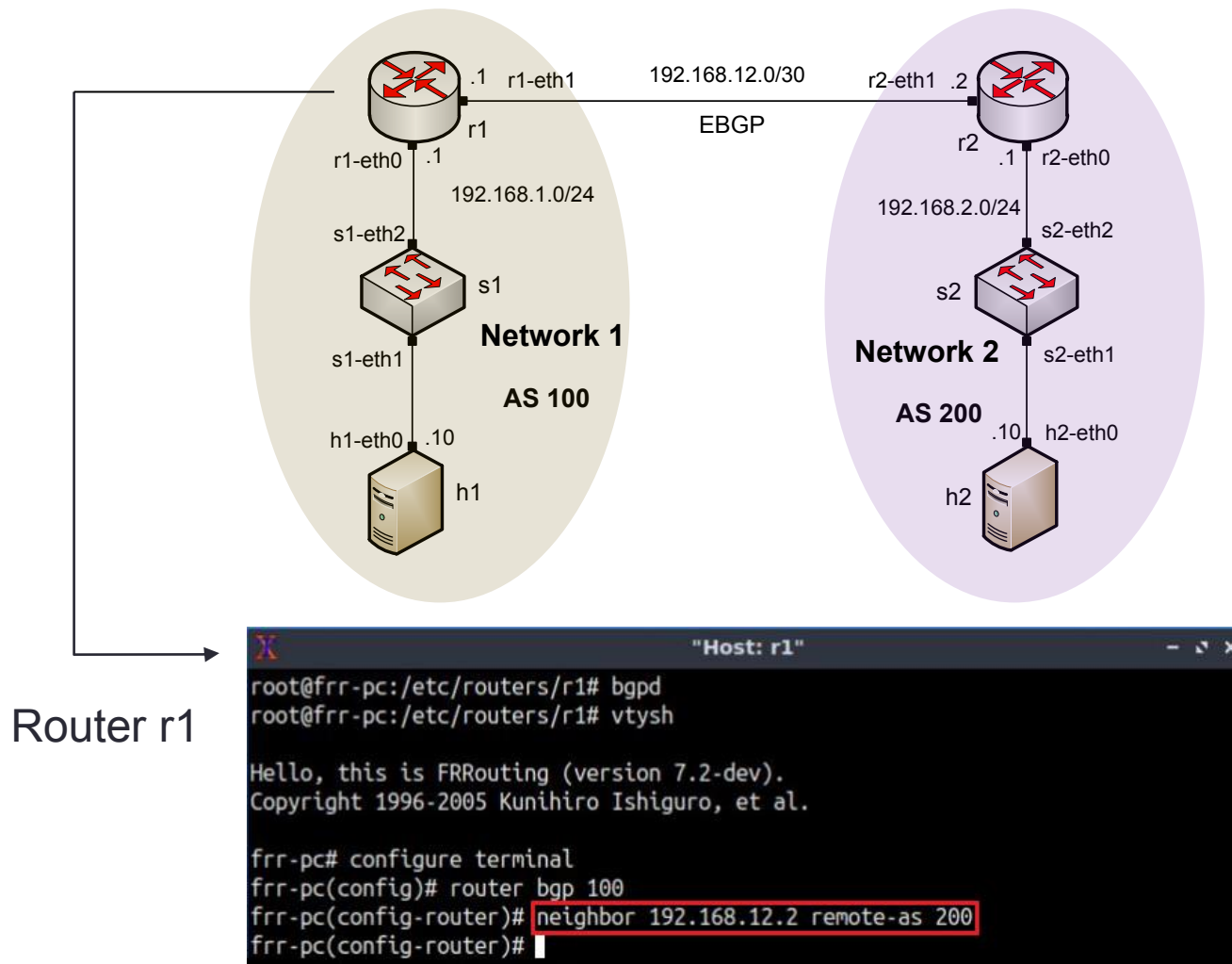
Lab 3 Topology

- Network 1 in AS 100 consists of a host, a switch, and a router
- Network 2 in AS 200 consists of a host, a switch, and a router



Lab 3 Configuration

- Establishing BGP neighborhood

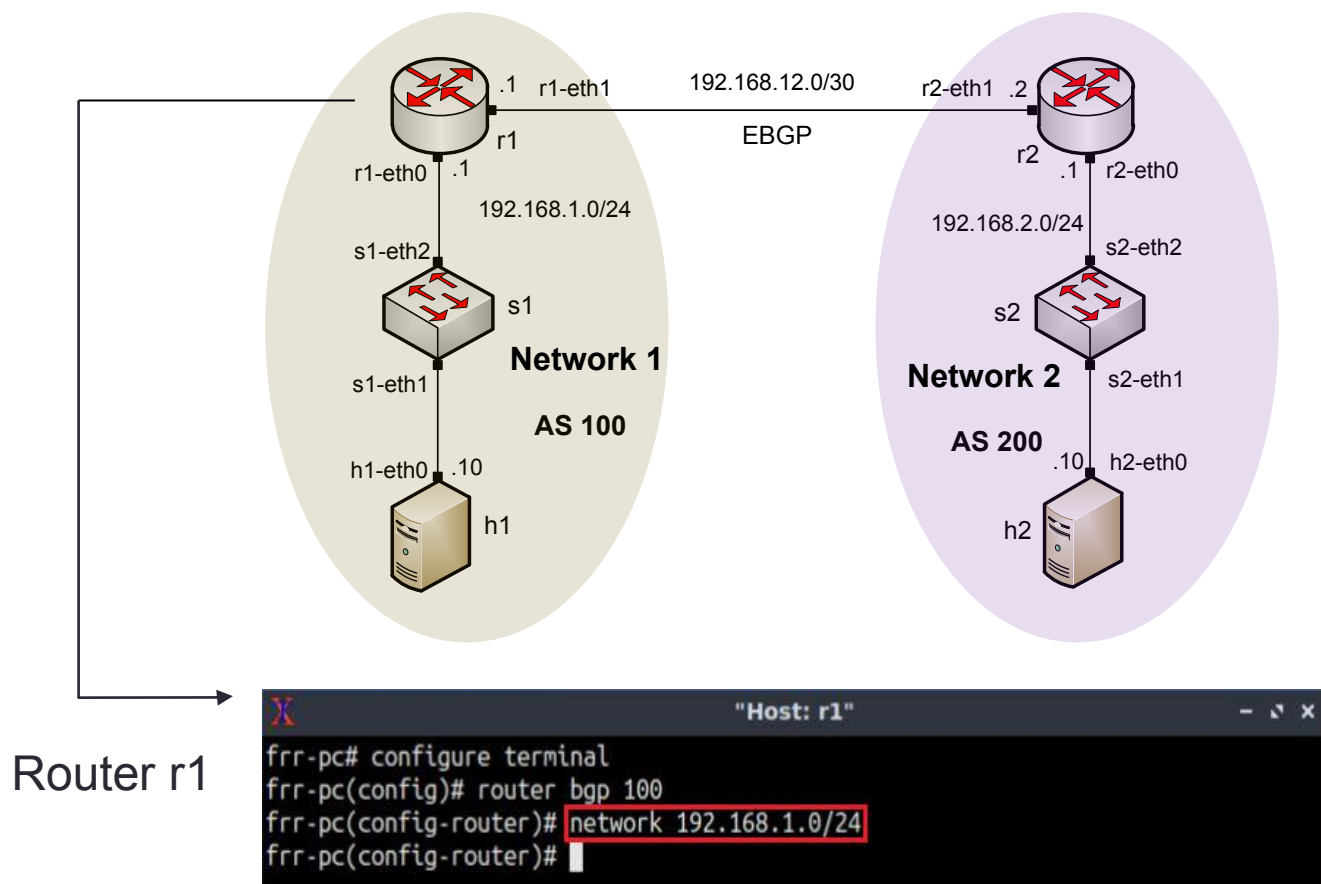


```
X "Host: r2"
frr-pc# ping 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data.
64 bytes from 192.168.12.1: icmp_seq=1 ttl=64 time=0.067 ms
64 bytes from 192.168.12.1: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 192.168.12.1: icmp_seq=3 ttl=64 time=0.034 ms
^C
--- 192.168.12.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 30ms
rtt min/avg/max/mdev = 0.034/0.047/0.067/0.016 ms
frr-pc#
```

```
X "Host: r2"
frr-pc# ping 192.168.1.10
connect: Network is unreachable
frr-pc#
```

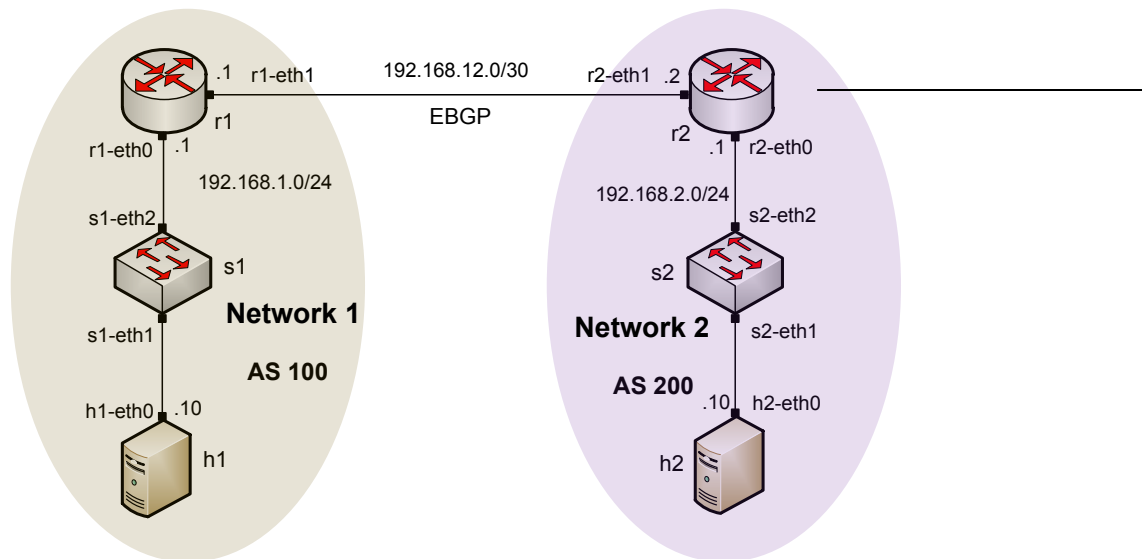
Lab 3 Configuration

- Advertising a network in BGP



Lab 3 Configuration

- Routing table: lists the routes learned from different routing protocols



```

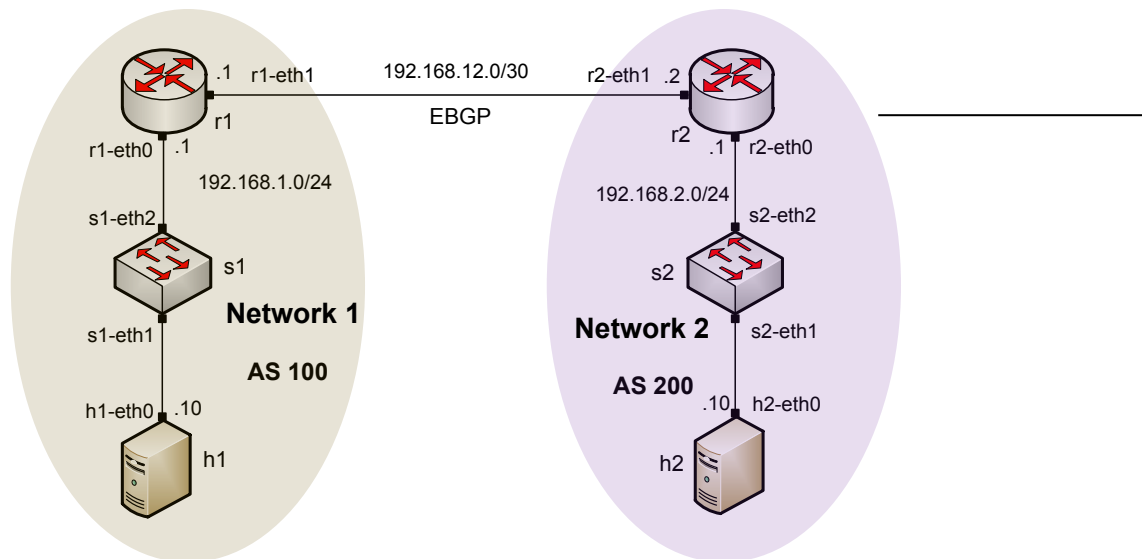
X "Host: r2"
frr-pc# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
F - PBR, f - OpenFabric,
> - selected route, * - FIB route, q - queued route, r - rejected route

B>* 192.168.1.0/24 [20/0] via 192.168.12.1, r2-eth1, 00:00:52
C>* 192.168.2.0/24 is directly connected, r2-eth0, 00:18:36
C>* 192.168.12.0/30 is directly connected, r2-eth1, 00:18:02
frr-pc#
  
```

Router r2

Lab 3 Configuration

- BGP table: lists the routes learned from BGP routing protocol



```

Host: r2
frr-pc# show ip bgp
BGP table version is 2, local router ID is 192.168.12.2, vrf id 0
Default local pref 100, local AS 200
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Next hop codes: @NNN next hop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 192.168.1.0/24  192.168.12.1      0             0 100 i
*> 192.168.2.0/24  0.0.0.0           0             32768 i

Displayed 2 routes and 2 total paths
frr-pc#
  
```

Router r2


```
X "Host: h2"
root@frr-pc:~# ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=60 time=0.136 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=60 time=0.110 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=60 time=0.115 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=60 time=0.105 ms
64 bytes from 192.168.1.10: icmp_seq=5 ttl=60 time=0.102 ms
64 bytes from 192.168.1.10: icmp_seq=6 ttl=60 time=0.089 ms
^C
--- 192.168.1.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 113ms
rtt min/avg/max/mdev = 0.089/0.109/0.136/0.017 ms
root@frr-pc:~#
```

Security issues

- A router may be talking to a wrong device
 1. Reroutes traffic to an intruder (advertise prefix)
 2. Pretends to be your friend (spoofing)
- Content/Advertisements can be modified in transit

TCP reset attacks

EBGP sessions run over TCP

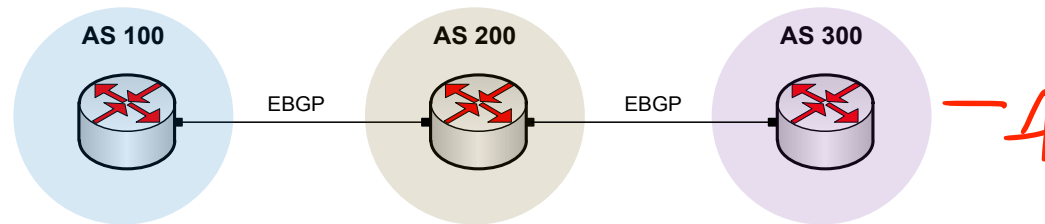


Figure 1. Routers in different ASes run EBGP to advertise routing information.

- If AS 100 sends TCP reset packet to AS 200, AS200 tear down TCP session with AS100
- An attacker can send a spoofed TCP reset to AS200 -- interrupts traffic flow between AS100 and AS200
- What an attacker needs?
 - right IP address, BGP port (179), right seq number
 - Simply sends packets with all possible sequence numbers – takes a minute or so

Security mechanism: TCP MD5 Hash Algorithm and password

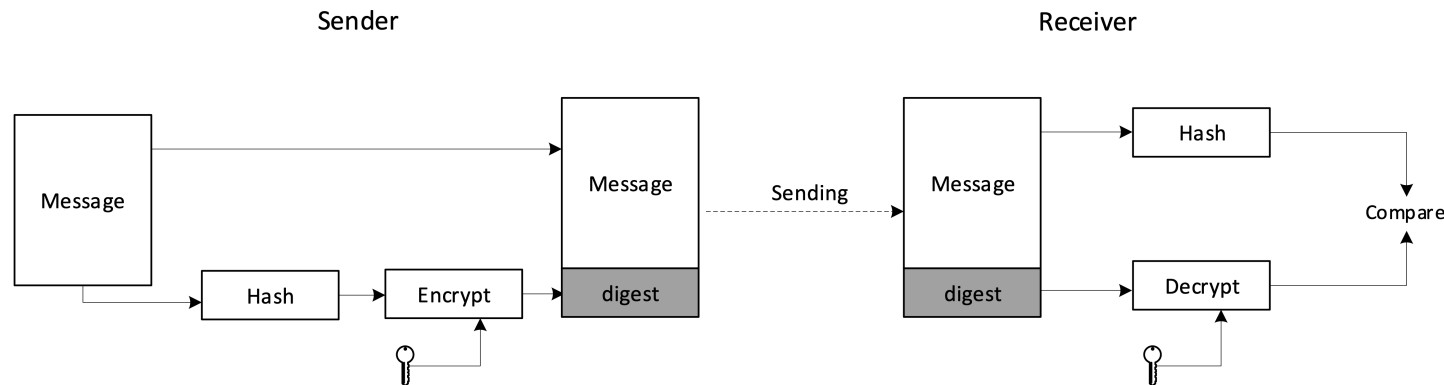
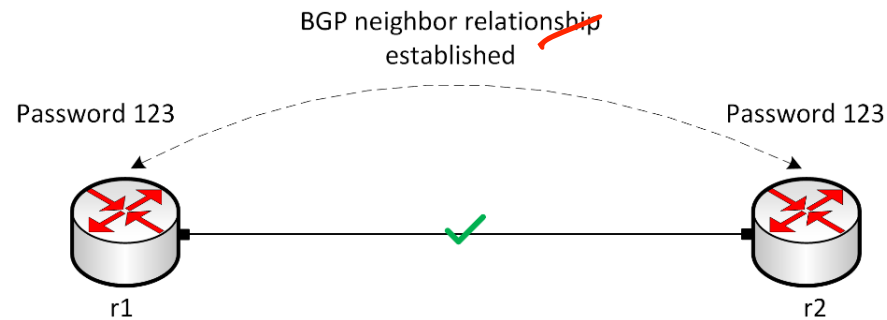


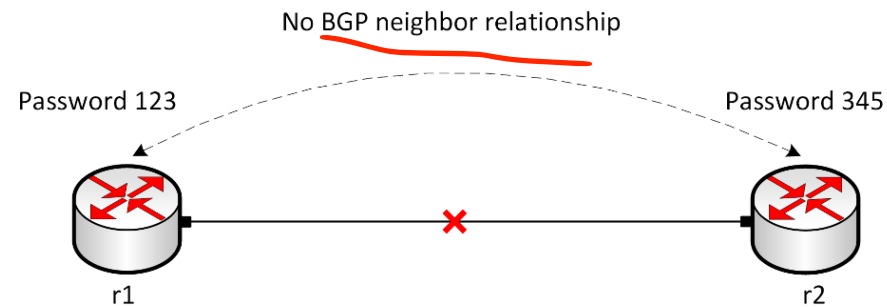
Figure 2. MD5 hash algorithm.

- TCP MD5 hash algorithm
 - Cryptographic function
 - Input: Arbitrary length message
 - Output: 128-bit message digest
- Computationally hard to produce
 - Two messages with same digest
 - Same message given a digest

BGP Authentication



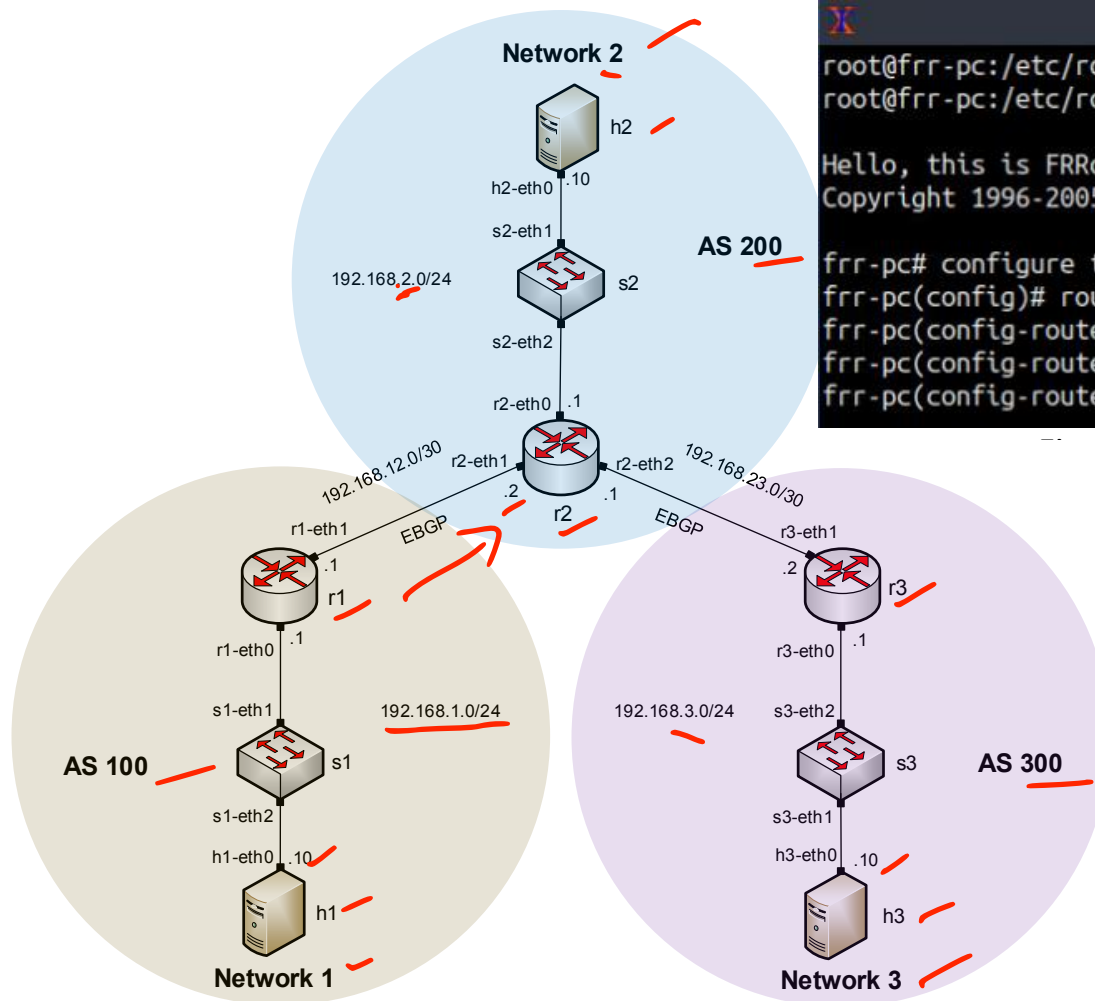
(a)



(b)

BGP peers must be configured with the same password to establish BGP neighbor relationship

Example topology

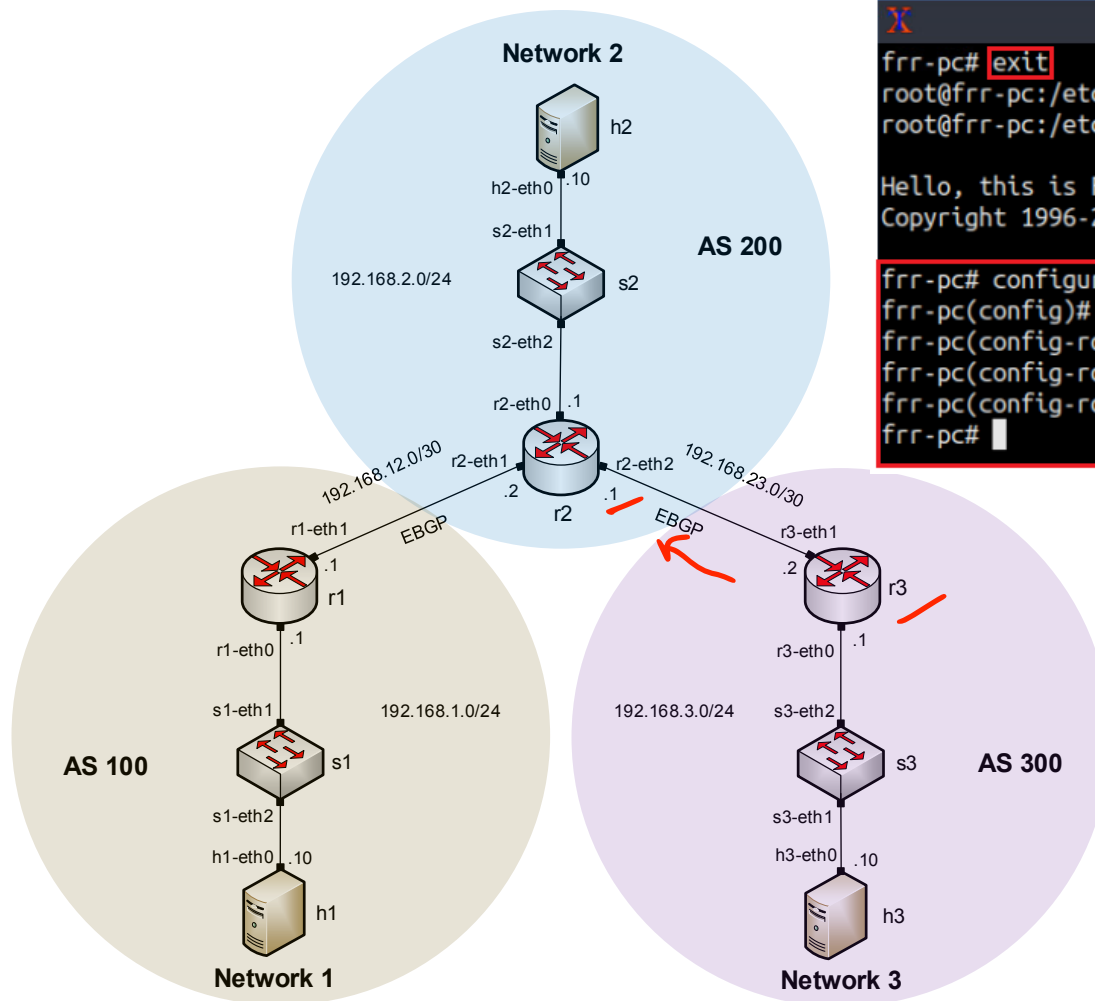


```
"Host: r1"
root@frr-pc:/etc/routers/r1# bgpd
root@frr-pc:/etc/routers/r1# vtysh

Hello, this is FRRouting (version 7.2-dev).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr-pc# configure terminal
frr-pc(config)# router bgp 100
frr-pc(config-router)# neighbor 192.168.12.2 remote-as 200
frr-pc(config-router)# network 192.168.1.0/24
frr-pc(config-router)#
```

Example topology

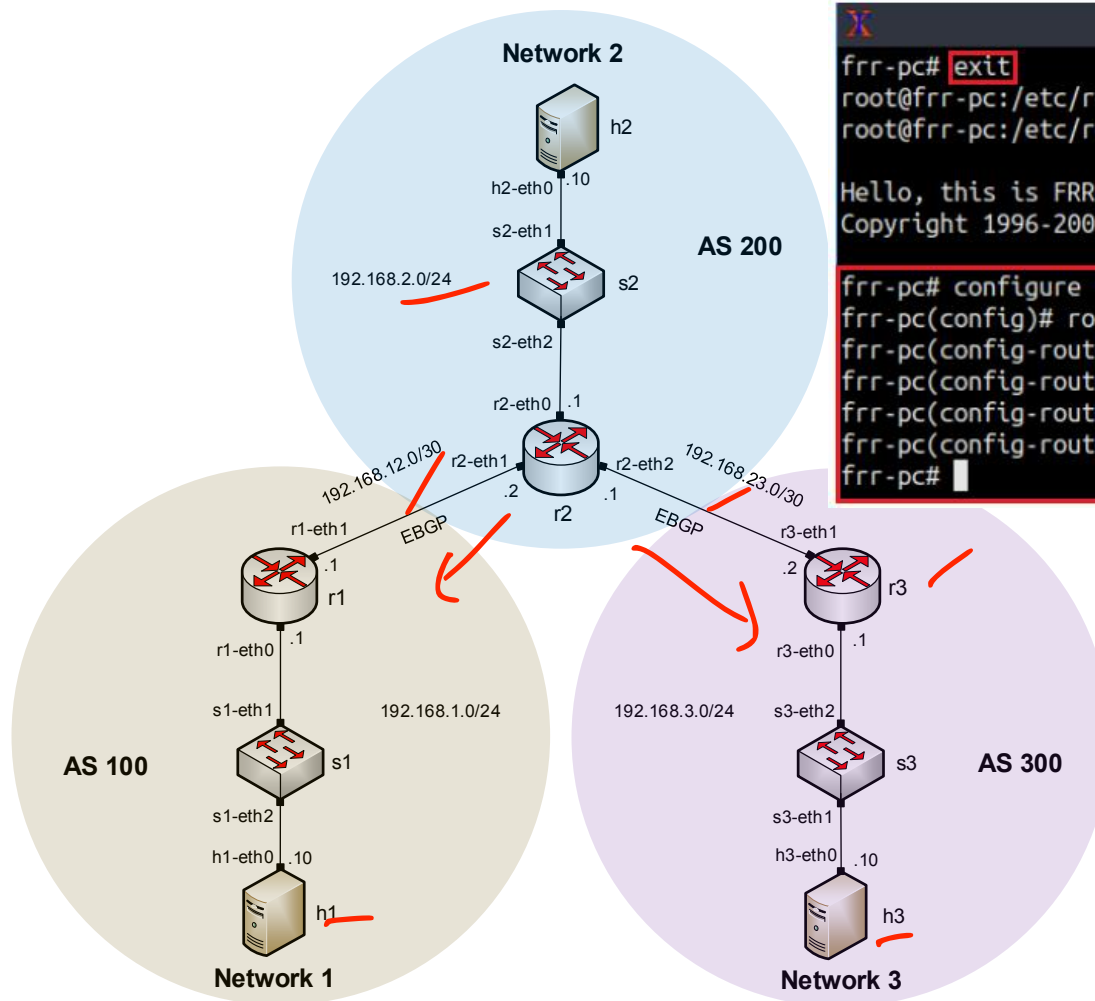


```
Host: r3
frr-pc# exit
root@frr-pc:/etc/routers/r3# bgpd
root@frr-pc:/etc/routers/r3# vtysh

Hello, this is FRRouting (version 7.2-dev).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr-pc# configure terminal
frr-pc(config)# router bgp 300
frr-pc(config-router)# neighbor 192.168.23.1 remote-as 200
frr-pc(config-router)# network 192.168.3.0/24
frr-pc(config-router)# end
frr-pc#
```

Example topology



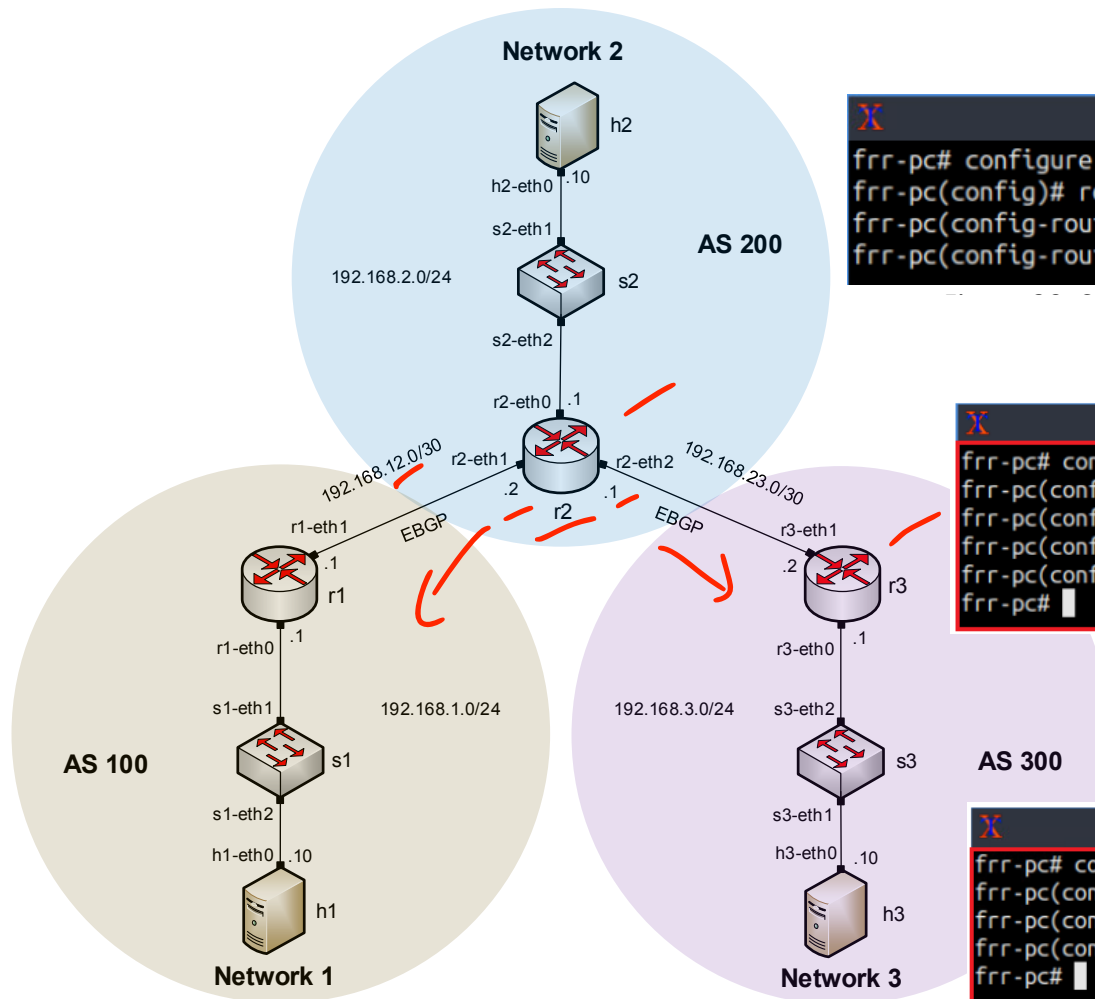
```
Host: r2
frr-pc# exit
root@frr-pc:/etc/routers/r2# bgpd
root@frr-pc:/etc/routers/r2# vtysh

Hello, this is FRRouting (version 7.2-dev).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr-pc# configure terminal
frr-pc(config)# router bgp 200
frr-pc(config-router)# neighbor 192.168.12.1 remote-as 100
frr-pc(config-router)# neighbor 192.168.23.2 remote-as 300
frr-pc(config-router)# network 192.168.2.0/24
frr-pc(config-router)# end
frr-pc#
```

```
Host: h3
root@frr-pc:~# ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data:
64 bytes from 192.168.1.10: icmp_seq=1 ttl=61 time=1.06 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=61 time=0.090 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=61 time=0.096 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=61 time=0.092 ms
^C
--- 192.168.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 45ms
rtt min/avg/max/mdev = 0.090/0.333/1.056/0.417 ms
root@frr-pc:~#
```


Set up password

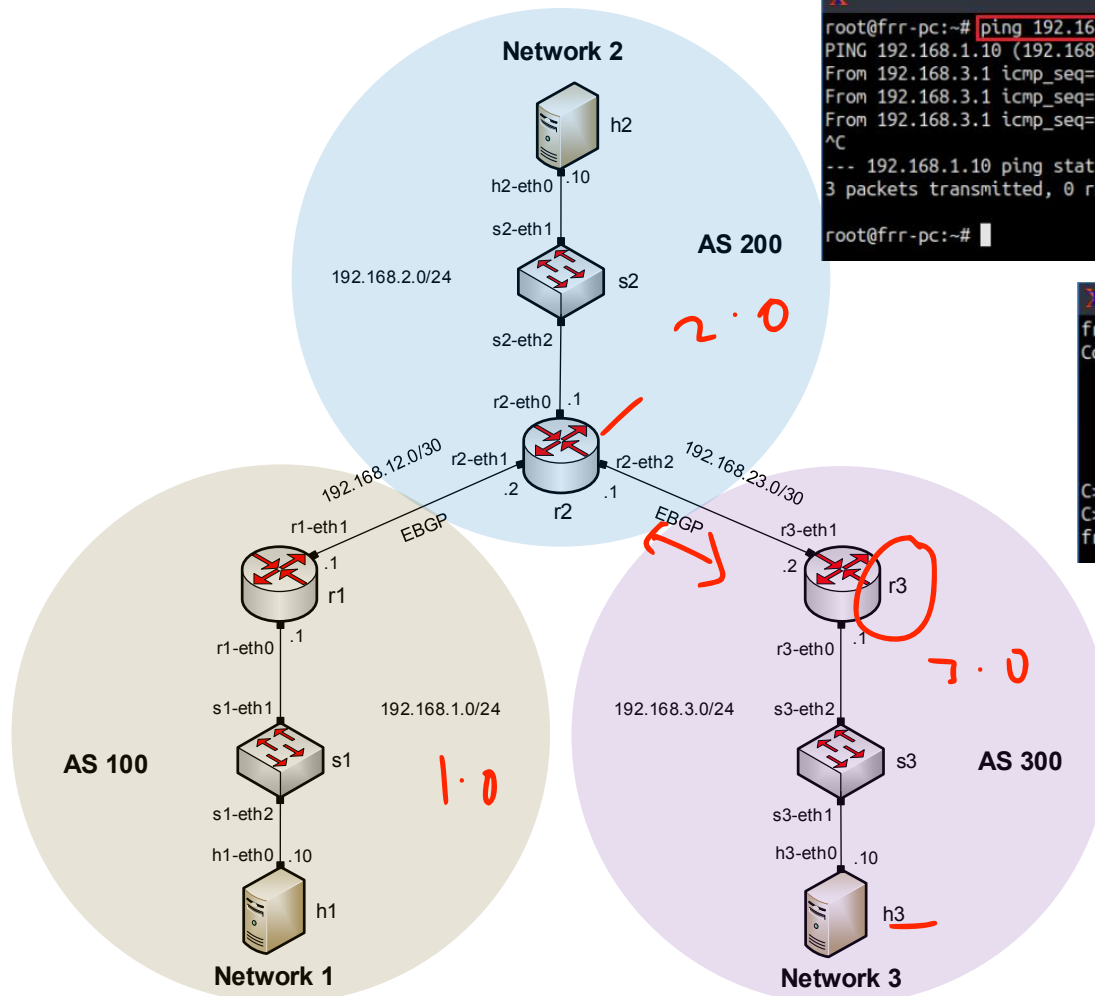


```
frr-pc# configure terminal
frr-pc(config)# router bgp 100
frr-pc(config-router)# neighbor 192.168.12.2 password 123
frr-pc(config-router)#
```

```
frr-pc# configure terminal
frr-pc(config)# router bgp 200
frr-pc(config-router)# neighbor 192.168.12.1 password 123
frr-pc(config-router)# neighbor 192.168.23.2 password 123
frr-pc(config-router)# end
frr-pc#
```

```
frr-pc# configure terminal
frr-pc(config)# router bgp 300
frr-pc(config-router)# neighbor 192.168.23.1 password 345
frr-pc(config-router)# end
frr-pc#
```

Verify MD5 authentication



```
root@frr-pc:~# ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data:
From 192.168.3.1 icmp_seq=1 Destination Net Unreachable
From 192.168.3.1 icmp_seq=2 Destination Net Unreachable
From 192.168.3.1 icmp_seq=3 Destination Net Unreachable
^C
--- 192.168.1.10 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 44ms
root@frr-pc:~#
```

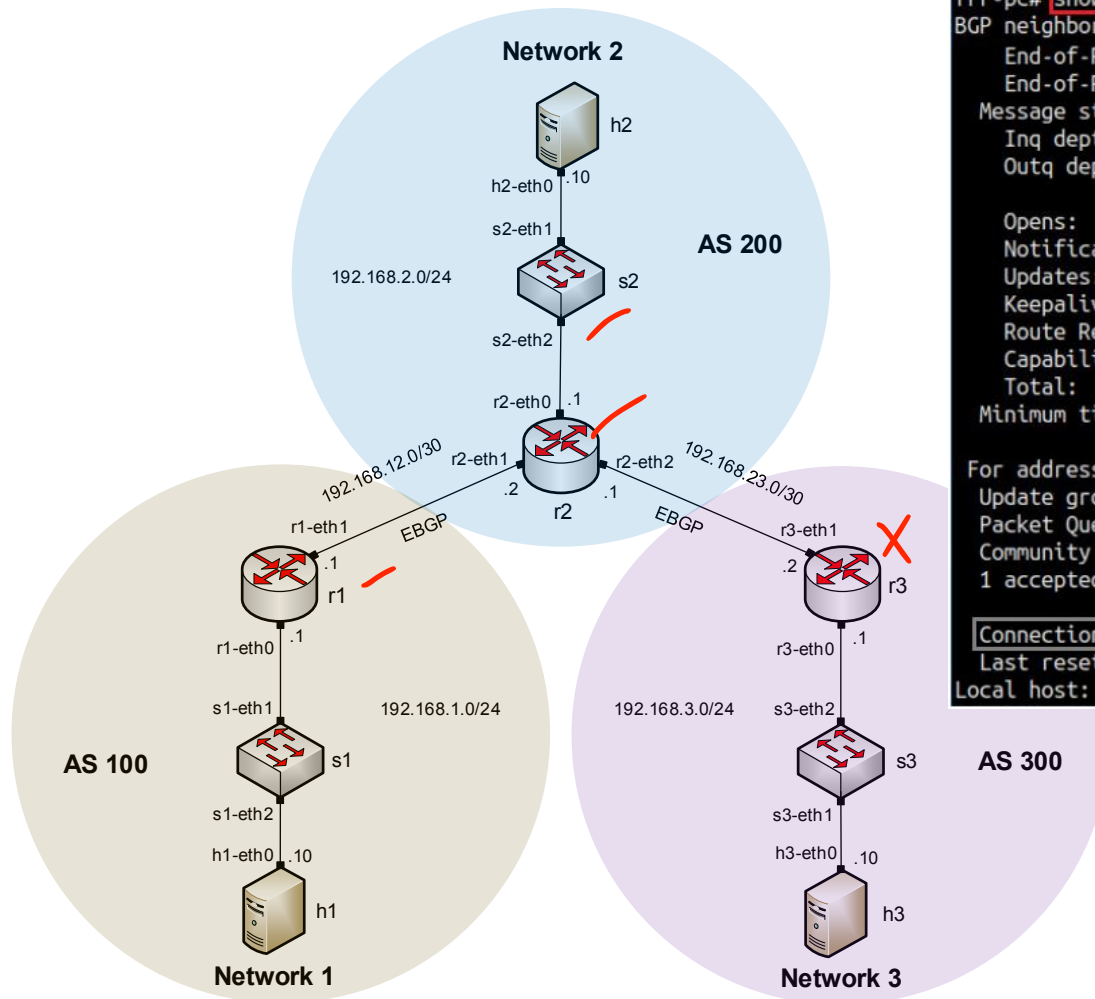
```
frr-pc# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

C>* 192.168.3.0/24 is directly connected, r3-eth0, 00:48:09
C>* 192.168.23.0/30 is directly connected, r3-eth1, 00:48:09
frr-pc#
```

```
frr-pc# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

B>* 192.168.1.0/24 [20/0] via 192.168.12.1, r2-eth1, 00:22:43
C>* 192.168.2.0/24 is directly connected, r2-eth0, 00:51:03
C>* 192.168.12.0/30 is directly connected, r2-eth1, 00:51:03
C>* 192.168.23.0/30 is directly connected, r2-eth2, 00:51:03
frr-pc#
```

Verify MD5 authentication



```
Host: r2
frr-pc# show ip bgp neighbors
BGP neighbor is 192.168.12.1, remote AS 100, local AS 200, external link
  End-of-RIB send: IPv4 Unicast
  End-of-RIB received: IPv4 Unicast
  Message statistics:
    Inq depth is 0
    Outq depth is 0

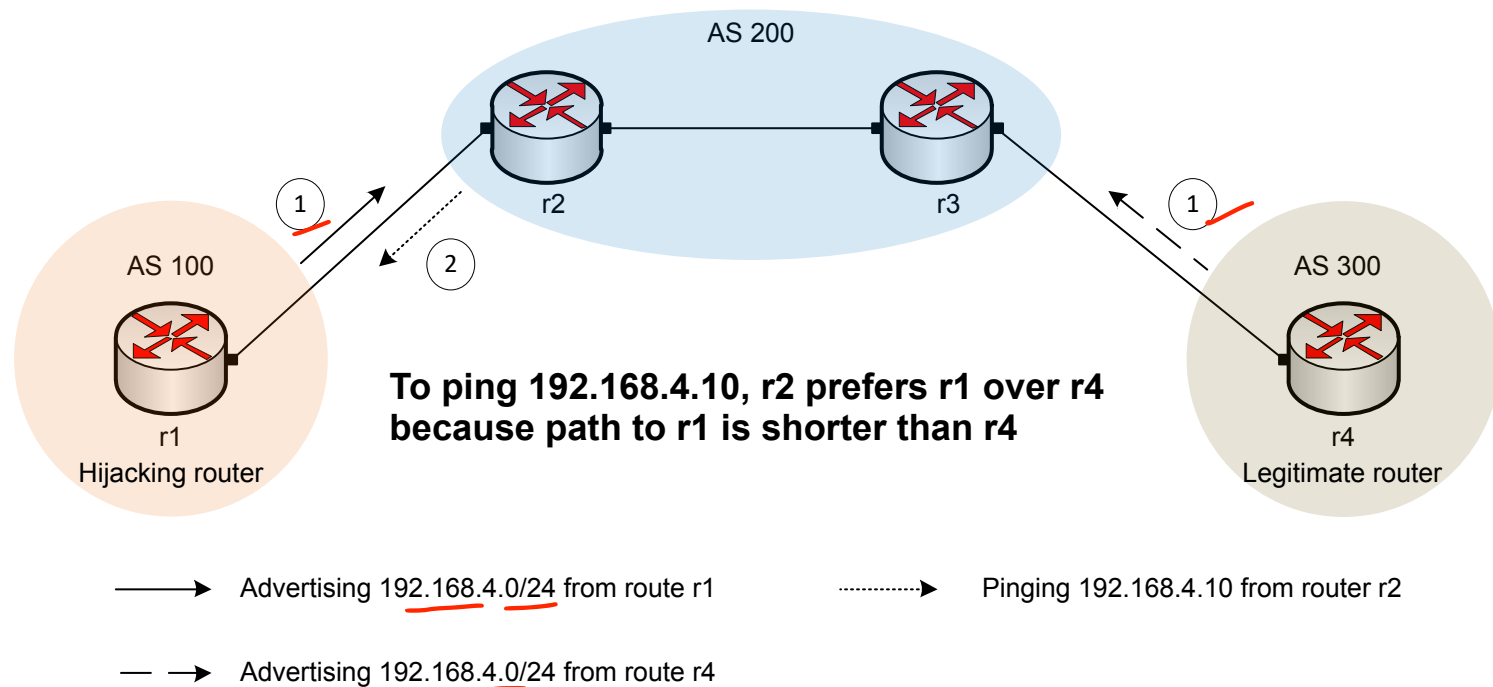
    Sent      Rcvd
  Opens:      2      2
  Notifications: 0      2
  Updates:    9      9
  Keepalives: 49     49
  Route Refresh: 0      0
  Capability: 0      0
  Total:     60     62
  Minimum time between advertisement runs is 0 seconds

  For address family: IPv4 Unicast
    Update group 1, subgroup 1
    Packet Queue length 0
    Community attribute sent to this neighbor(all)
    1 accepted prefixes

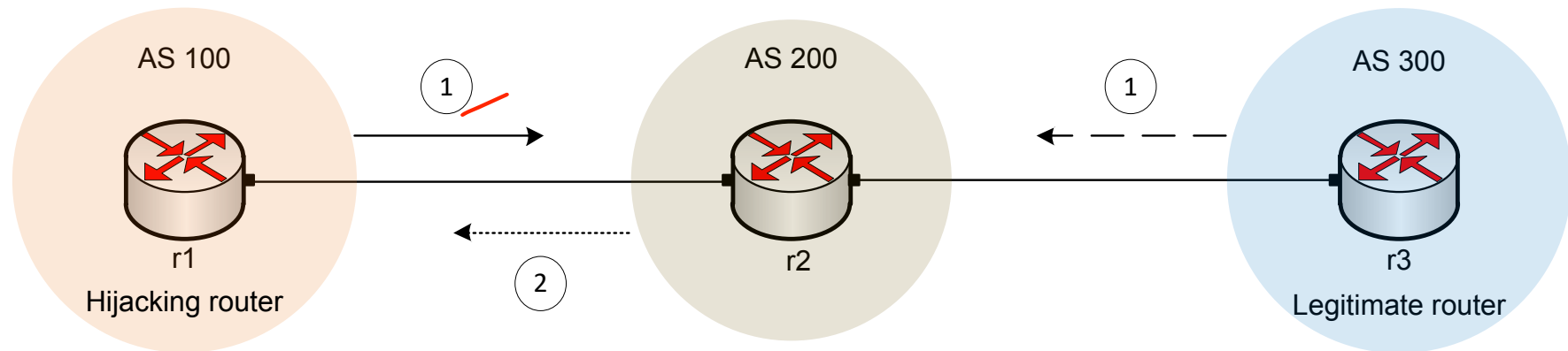
  Connections established 2; dropped 1
  Last reset 00:28:03, No AFI/SAFI activated for peer
  Local host: 192.168.12.2, Local port: 179
```

What is BGP Hijacking?

- BGP hijacking is when the attackers maliciously reroute Internet traffic
- It occurs when an unauthorized network originates IP prefix owned by other networks



BGP hijacking using specific IP prefix advertisement



—————> Advertising 192.168.3.0/25 from router r1

.....> Pinging 192.168.3.10 from router r2

— —> Advertising 192.168.3.0/24 from router r3

To ping the network 192.168.3.10, r2 prefers r1 over r3,
because r1 made specific announcement (/25)

- <https://www.bgpmon.net/large-scale-bgp-hijack-out-of-india/>
- <https://bgpstream.com/event/270621>

BGP Hijacking Attacks

- Large scale BGP hijack out of India (2015)¹
 - 16,123 hijacked prefixes
- BGP hijack affected Amazon DNS (2018)² ✓
 - 5 Amazon routes (prefixes) were affected
- Chinese Telecom performed a two hour BGP hijacking attack on European networks (2019)³
 - A significant portion of the traffic was routed through the Chinese Telecom infrastructure before reaching its destination
- Russian telecommunication provider rerouted traffic intended for several networks across the globe (2020)⁴
 - Over 8000 prefixes were rerouted from Cloudflare, Facebook, Google, Amazon, etc.

¹ Toonk, Andree "Large scale BGP hijack out of India". www.bgpmon.net/massive-route-leak-cause-internet-slowdown/

² Nichols Shaun, "AWS DNS network hijack turns MyEtherWallet into ThievesEtherWallet",

³ Linssen, R. H. H. G. M. "Vulnerability of DNS name servers against BGP hijacking." Bachelor's thesis, University of Twente, 2020

⁴ Improtta Alessandro, Sani Luca "April Fools' BGP Hijack". <https://blog.catchpoint.com/2020/04/06/april-fools-bgp-hijack/>

BGP Hijack 1: Thieves stole \$150K in Ethereum

- eNet was commandeered by miscreants to persuade its peers – potentially Hurricane Electric, Level 3, and others
- This reroute the internet's traffic from some AWS Route 53 DNS servers to a malicious DNS server acted as AWS DNS (hacked by thieves)
- The DNS servers misdirected visitors to MyEtherWallet.com to a phishing website
- How to stop?
 - DNSSEC = resolvers would deny fake records
 - HSTS = browsers would prevent visiting self-signed certs

source: https://www.theregister.com/2018/04/24/myetherwallet_dns_hijack/

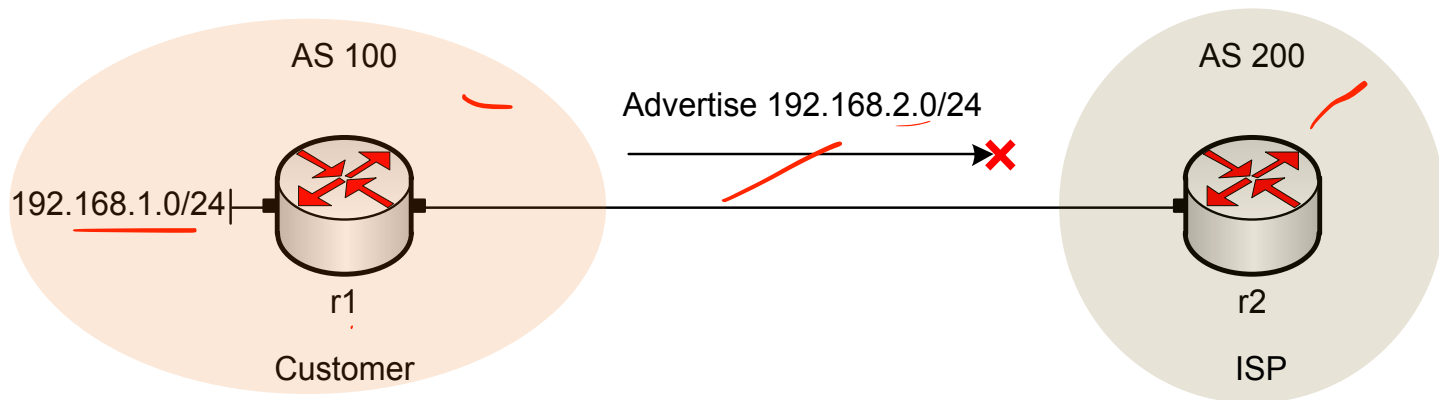
BGP Hijack 2: China telecom routes European traffic

- Swiss company Safe Host leaking over 70,000 routes to China Telecom in Germany.
- The Chinese telecommunication company then announced these routes on to the global Internet, which resulted in large amounts of web traffic destined for some of the largest European mobile networks to be redirected through China Telecom's network.
- In 2010, China Telecom hijacked 15% of the world's Internet prefixes, which resulted in popular websites being rerouted through China for around 18 minutes. The incident impacted US government (".gov") and military (".mil") sites as well..
- The carrier has been constantly misdirecting Internet traffic through its network in China for several years.
- How to stop ? -- RPKI to secure BGP

source: <https://www.securityweek.com/china-telecom-routes-european-traffic-its-network-two-hours>,
https://essay.utwente.nl/80731/1/Linssen_BA_EEMCS.pdf

Using IP Prefix Filters to Mitigate BGP Hijacking

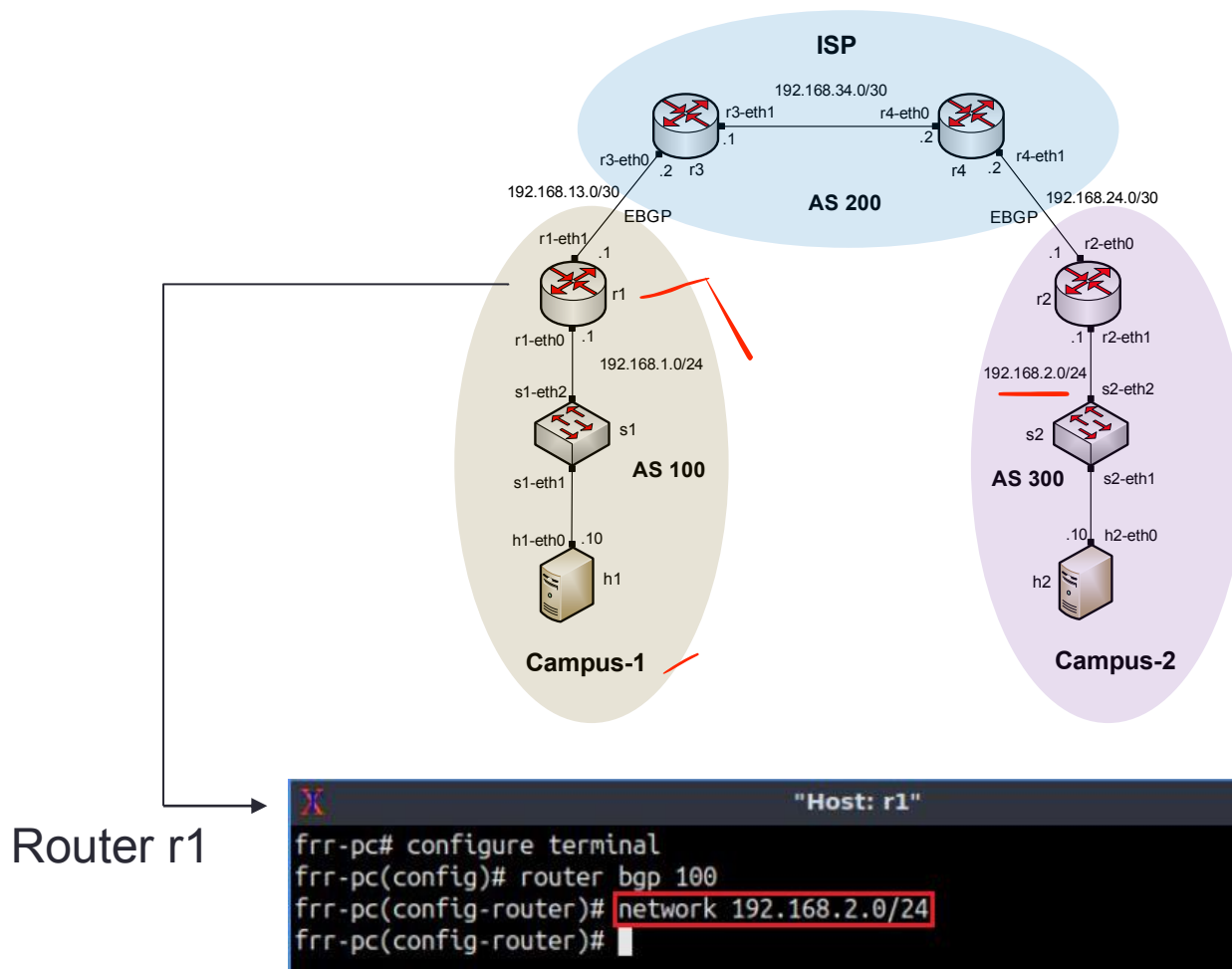
- A router can limit the number of BGP route advertisements by configuring IP prefix filters
- “Most important is to secure the inbound routing advertisements, particularly from customer networks, through the use of explicit prefix-level filters...”¹



¹ “MANRS Implementation Guide”. <https://www.manrs.org/isps/guide/filtering/>

Lab 13 Configuration

- Router r1 hijacks (advertises) the network 192.168.2.0/24



Lab 13 Configuration

- Router r3 changes the next hop of Campus-2 (192.168.2.0/24)

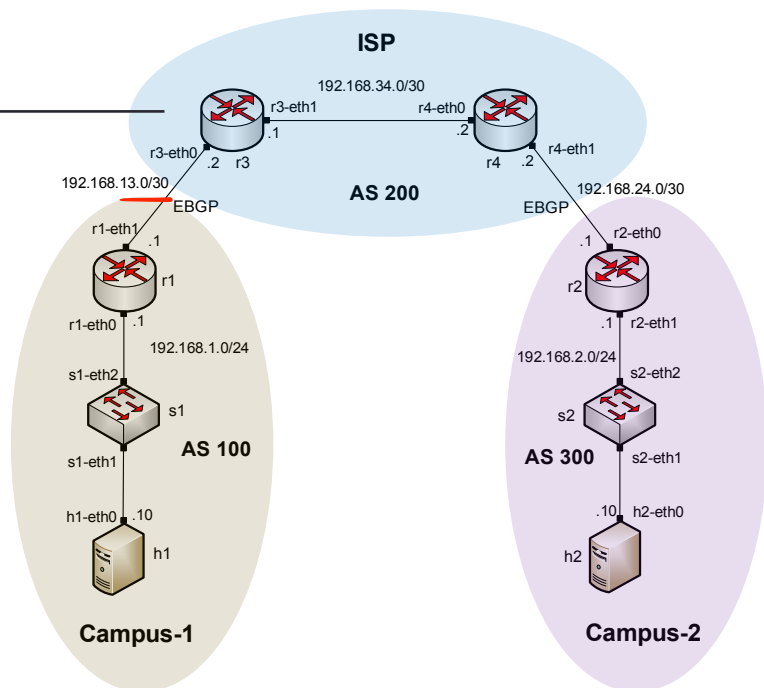
Router r3

```

Host: r3
frr-pc# show ip bgp
BGP table version is 3, local router ID is 192.168.34.1, vrf id 0
Default local pref 100, local AS 200
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

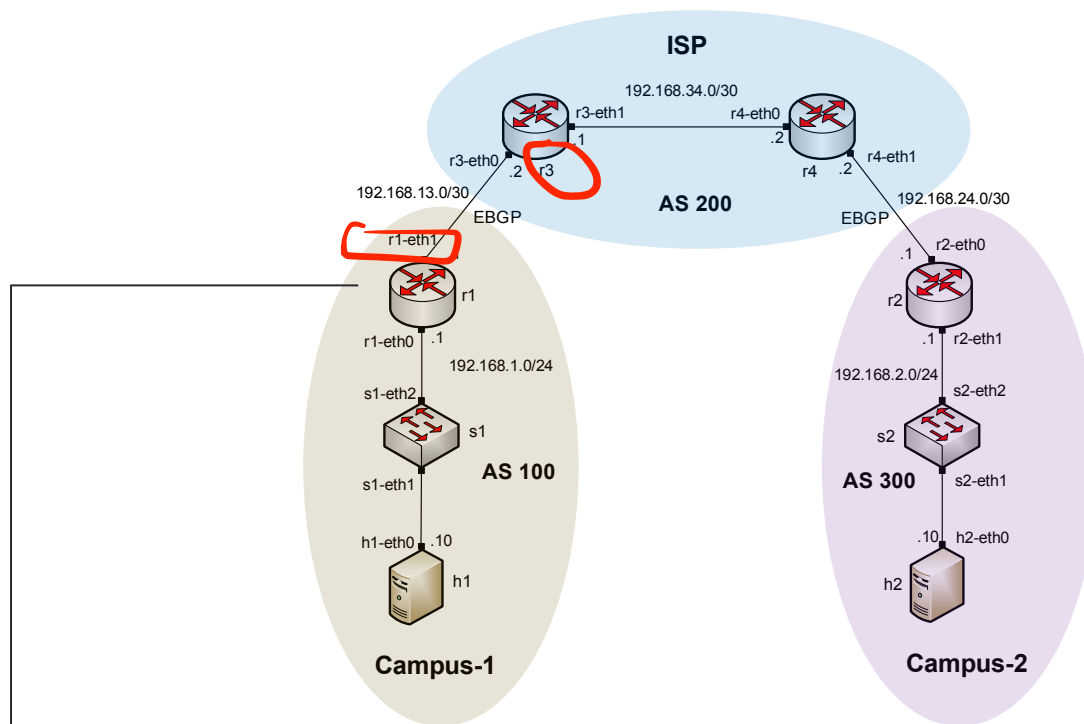
   Network        Next Hop           Metric LocPrf Weight Path
*> 192.168.1.0/24  192.168.13.1         0           0 100 i
*> 192.168.2.0/24  192.168.13.1         0           0 100 i
*i               192.168.34.2         0          100 0 300 i

Displayed 2 routes and 3 total paths
frr-pc#
  
```



Lab 13 Configuration

- Capture the packets on router r1, specifically at r1-eth1



Router r1

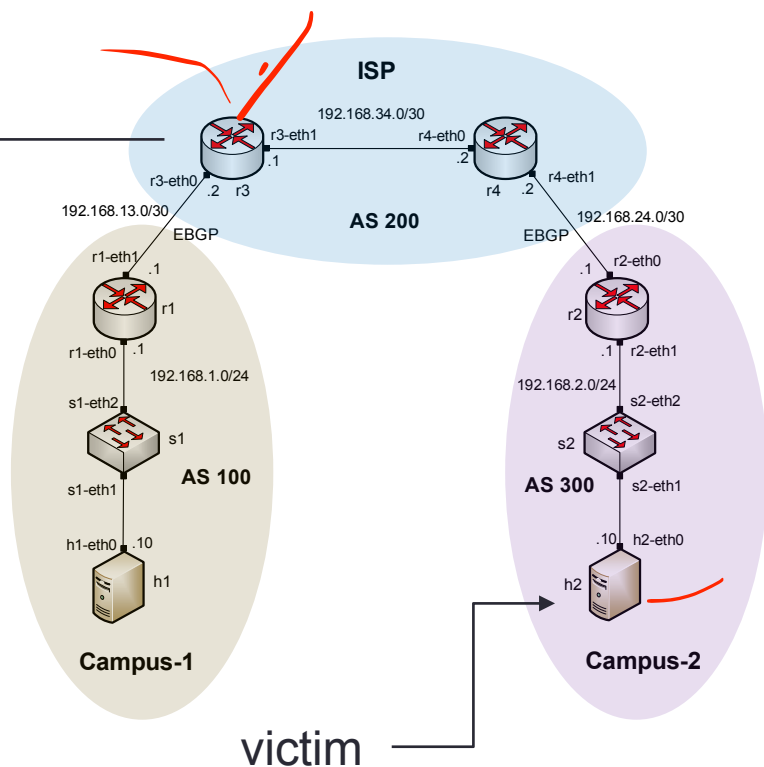
```
"Host: r1"
root@frr-pc:/etc/routers/r1# tcpdump -i r1-eth1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on r1-eth1, link-type EN10MB (Ethernet), capture size 262144 bytes
```

Lab 13 Configuration

- Ping the victim (192.168.2.10) from the ISP (router r3)

Router r3

```
frr-pc# ping 192.168.2.10
PING 192.168.2.10 (192.168.2.10) 56(84) bytes of data.
From 192.168.13.1 icmp_seq=1 Destination Net Unreachable
From 192.168.13.1 icmp_seq=2 Destination Net Unreachable
From 192.168.13.1 icmp_seq=3 Destination Net Unreachable
From 192.168.13.1 icmp_seq=4 Destination Net Unreachable
^C
--- 192.168.2.10 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 76ms
frr-pc#
```



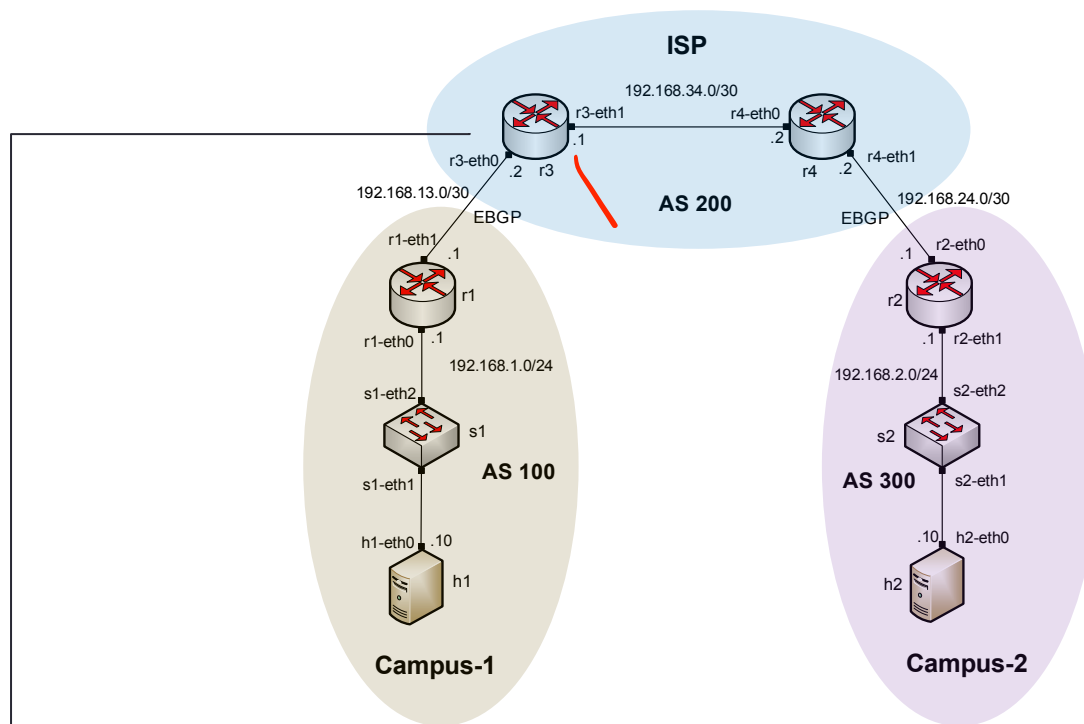
Lab 13 Configuration

- The traffic to network 192.168.2.10 will be rerouted to the hijacking router

```
"Host: r1"
16:42:56.954553 IP 192.168.13.1 > 192.168.13.2: ICMP net 192.168.2.10 unreachable, length 92
16:42:57.978489 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 3, length 64
16:42:57.978522 IP 192.168.13.1 > 192.168.13.2: ICMP net 192.168.2.10 unreachable, length 92
16:42:59.002492 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 4, length 64
16:42:59.002529 IP 192.168.13.1 > 192.168.13.2: ICMP net 192.168.2.10 unreachable, length 92
16:43:00.026860 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 5, length 64
16:43:01.050427 ARP, Request who-has 192.168.13.1 tell 192.168.13.2, length 28
16:43:01.050605 ARP, Reply 192.168.13.1 is-at da:e0:e3:9f:dd:c9 (oui Unknown), length 28
16:43:01.050577 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 6, length 64
16:43:02.074485 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 7, length 64
16:43:03.098485 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 8, length 64
16:43:04.122487 IP 192.168.13.2 > 192.168.2.10: ICMP echo request, id 2045, seq 9, length 64
```

Lab 13 Configuration

- Configure IP prefix list on the ISP (router r3)



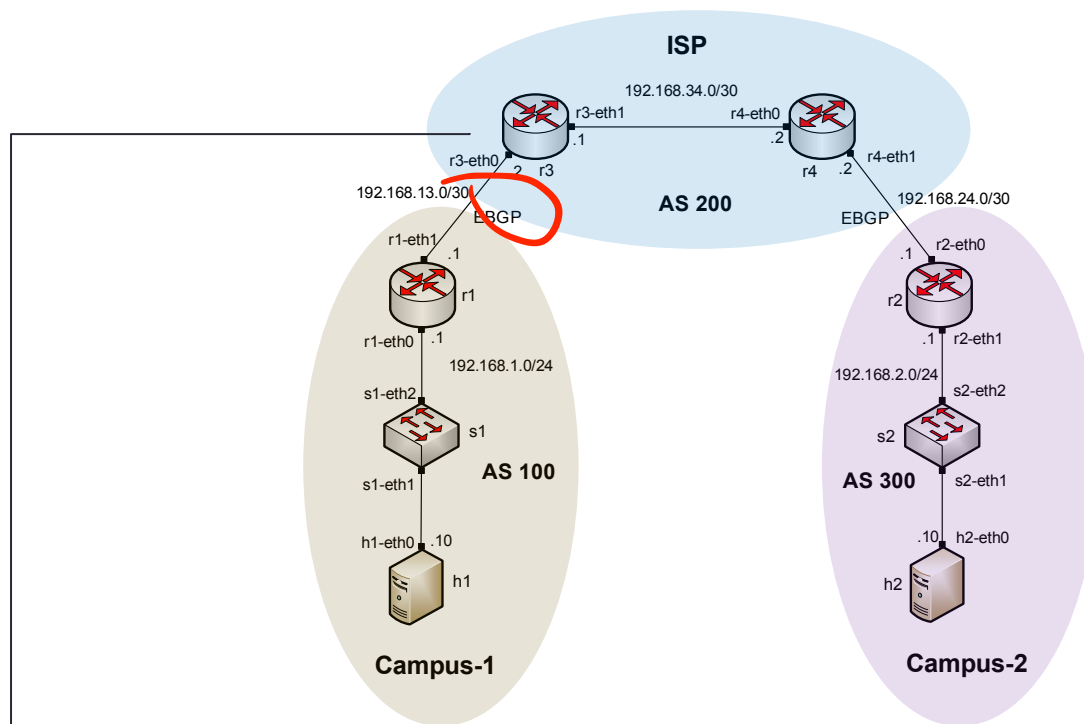
Router r3

```

"Host: r3"
frr-pc# configure terminal
frr-pc(config)# ip prefix-list campus1-in seq 10 permit 192.168.1.0/24
frr-pc(config)# router bgp 200
frr-pc(config-router)#
  
```

Lab 13 Configuration

- Apply the prefix list to router r3 neighbor



Router r3

```

"Host: r3"
frr-pc# configure terminal
frr-pc(config)# ip prefix-list campus1-in seq 10 permit 192.168.1.0/24
frr-pc(config)# router bgp 200
frr-pc(config-router)# neighbor 192.168.13.1 prefix-list campus1-in in
frr-pc(config-router)#
  
```


Lab 13 Configuration

- Router r3 readjusts its BGP table back to normal

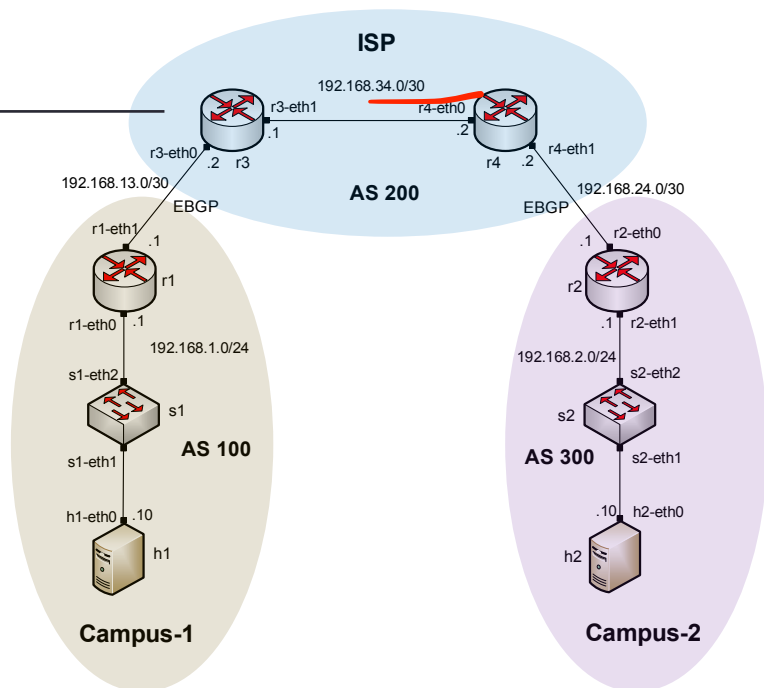
Router r3

```

Host: r3
frr-pc# show ip bgp
BGP table version is 4, local router ID is 192.168.34.1, vrf id 0
Default local pref 100, local AS 200
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

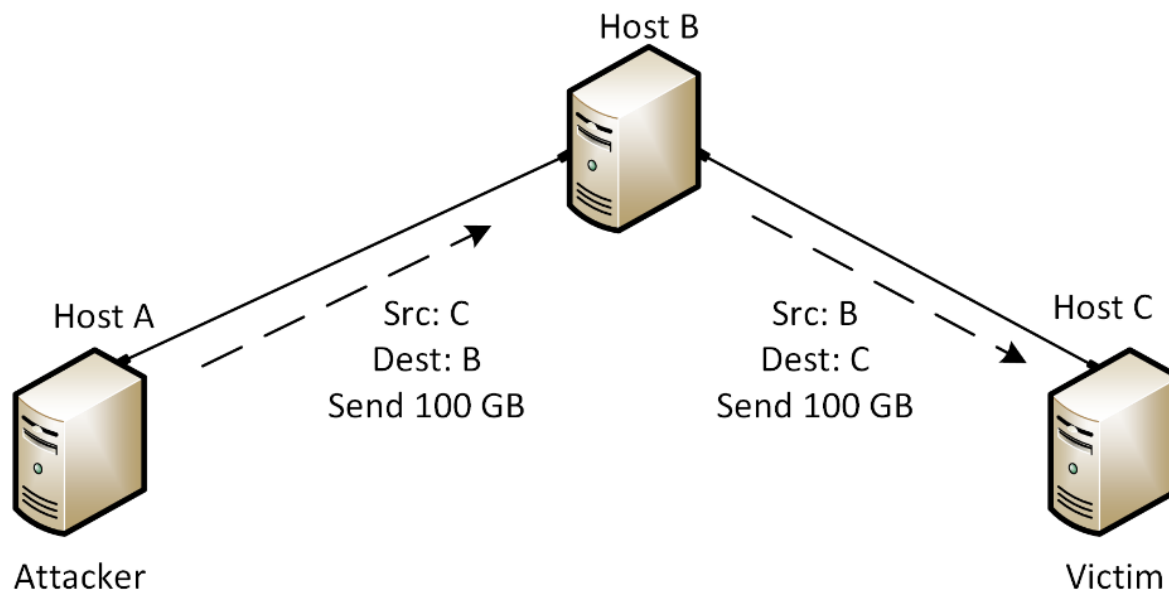
   Network        Next Hop                Metric LocPrf Weight Path
*> 192.168.1.0/24  192.168.13.1              0         0 100 i
*>i192.168.2.0/24  192.168.34.2              0        100   0 300 i

Displayed 2 routes and 2 total paths
frr-pc#
  
```



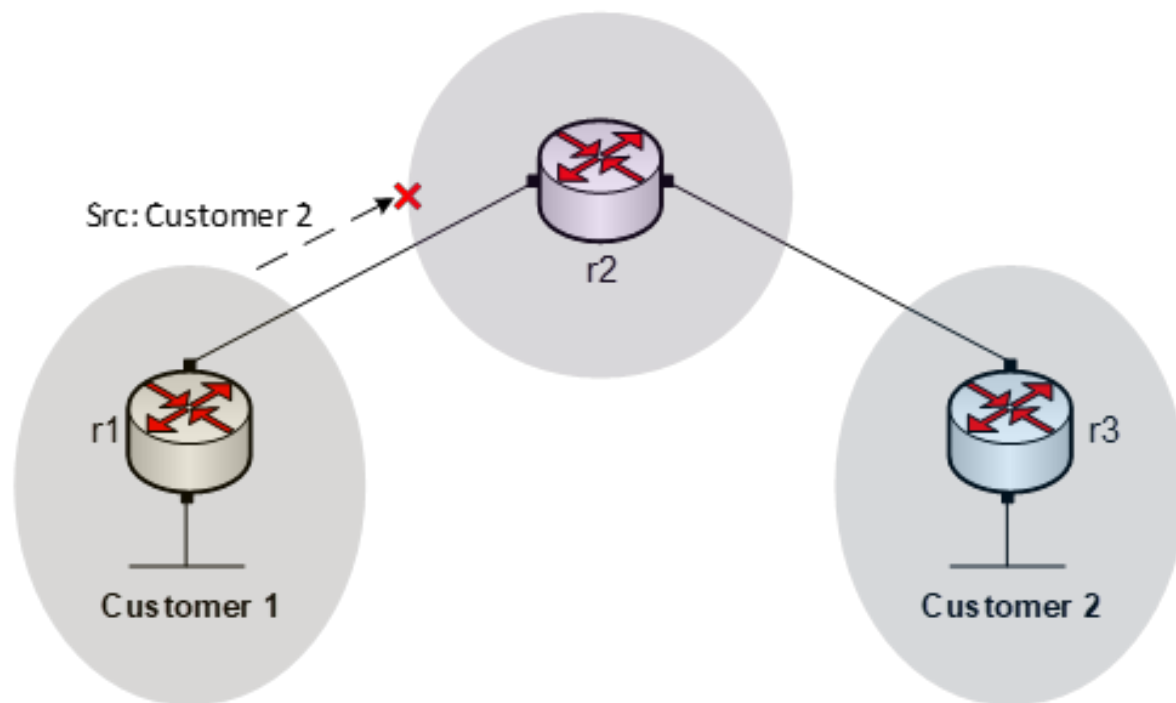
What is IP Spoofing?

- It is the process of originating IP packets with source addresses other than those assigned to the origin host.
- IP spoofing can be exploited in several ways, mainly to launch Denial of Service (DoS) attacks.



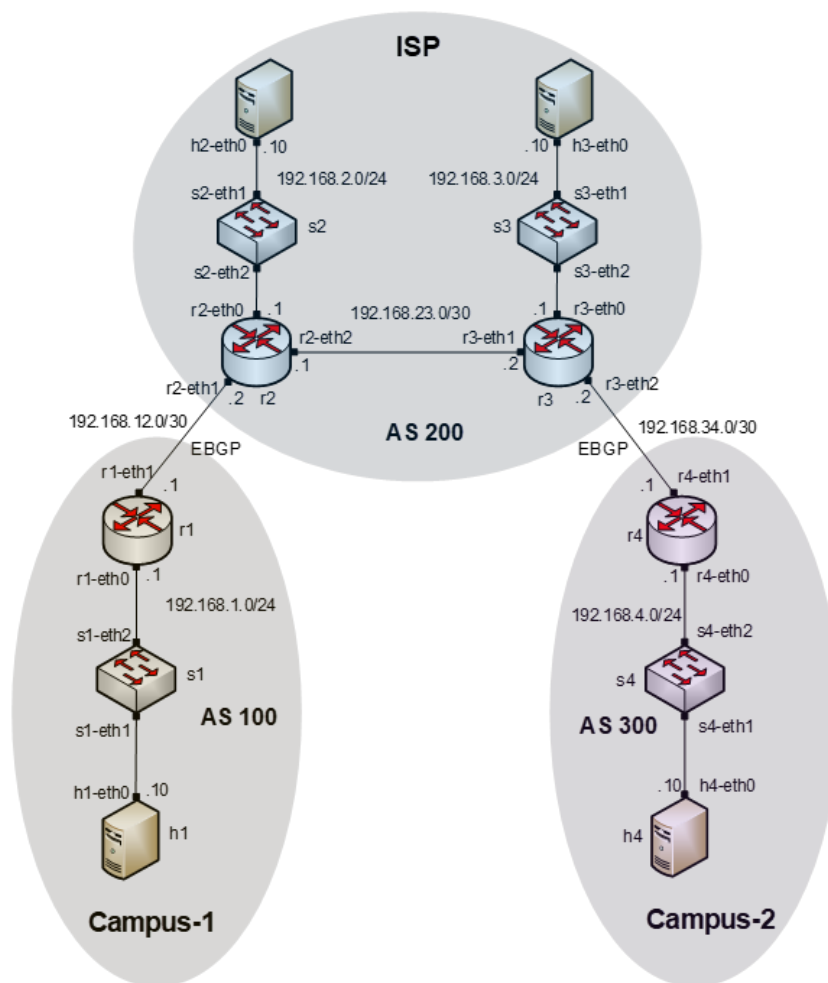
Anti-Spoofing Techniques – Route Filtering

- Route filtering is a method for selectively identifying routes that are advertised or received from neighbor routers.
- It can be used to manipulate traffic flows, reduce memory utilization, or to improve security



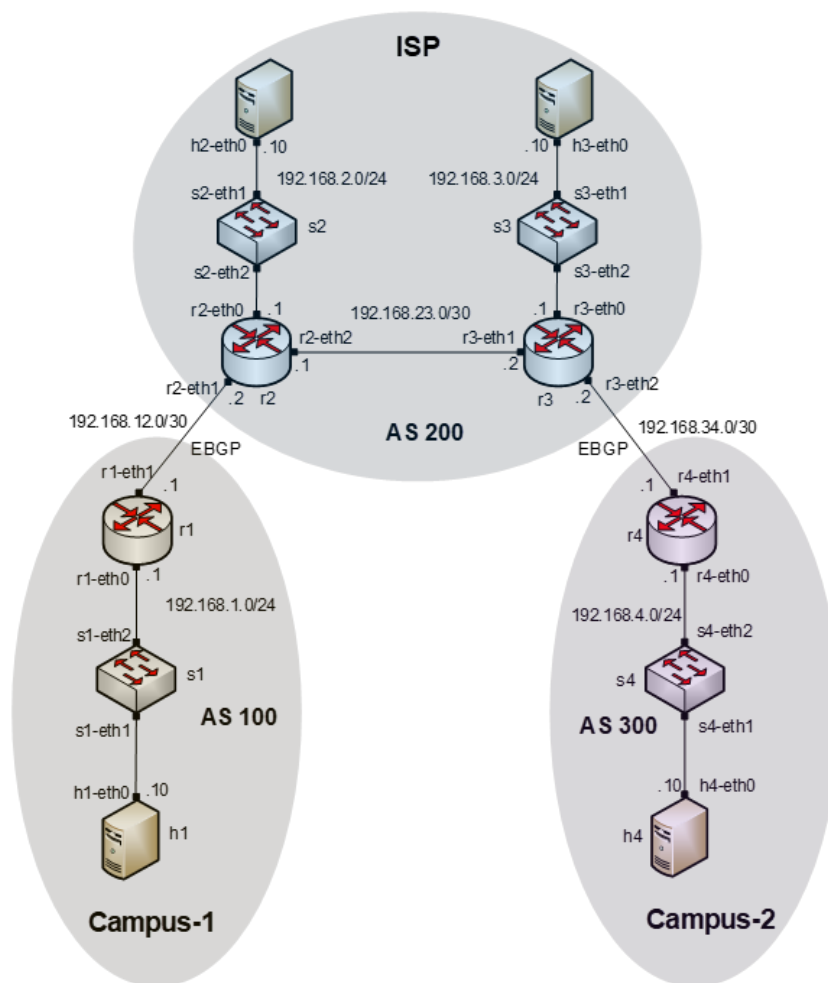
Lab 12 Topology

- Host h1 in Campus-1 spoofs the IP address of host h4 in Campus-2 and launches DoS attack.



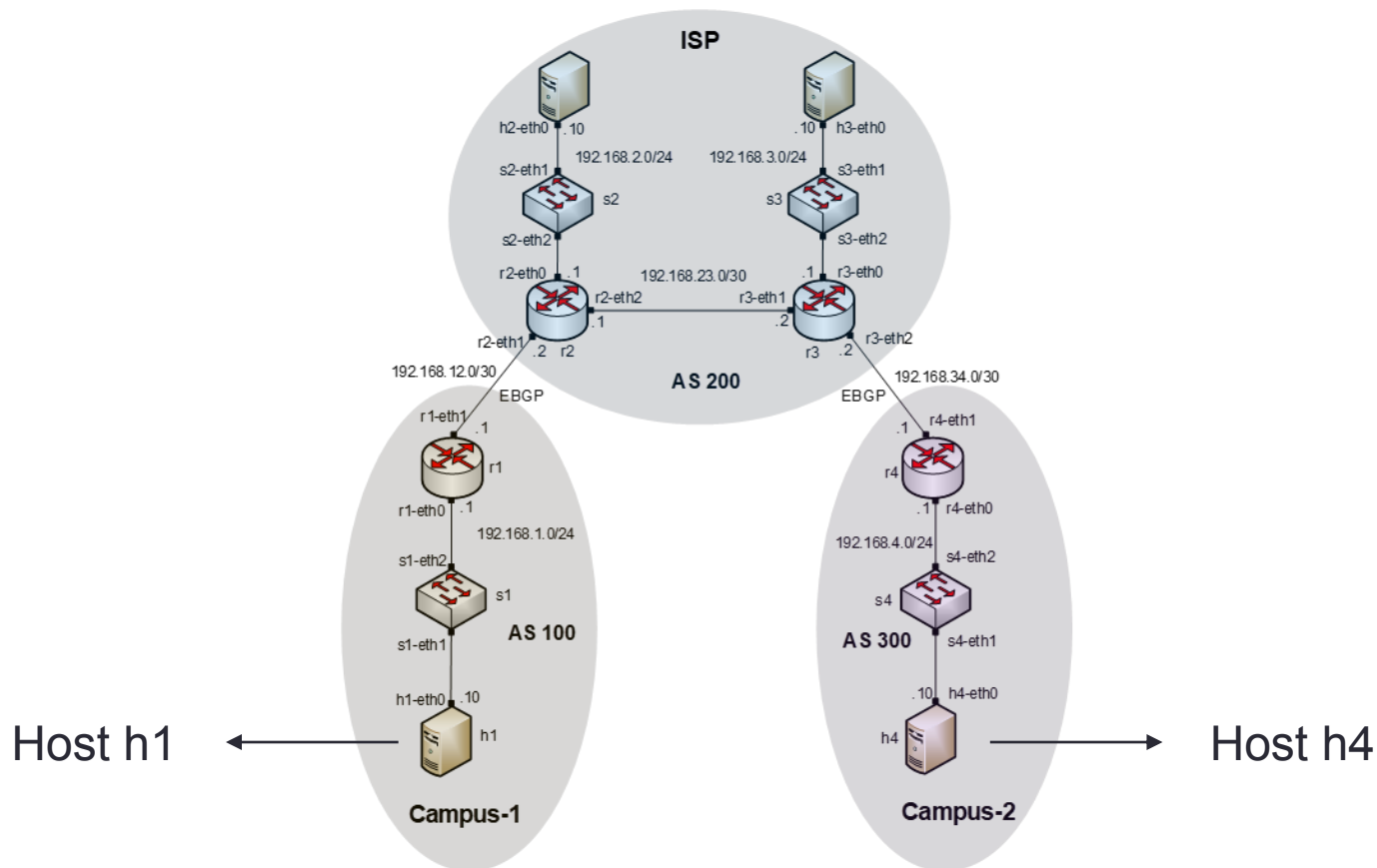
Lab 12 Topology

- The Internet Service Provider (ISP) applies the appropriate route filters to prevent IP spoofing



Lab 12 Configuration

- Host h1 spoofs the IP address of host h4

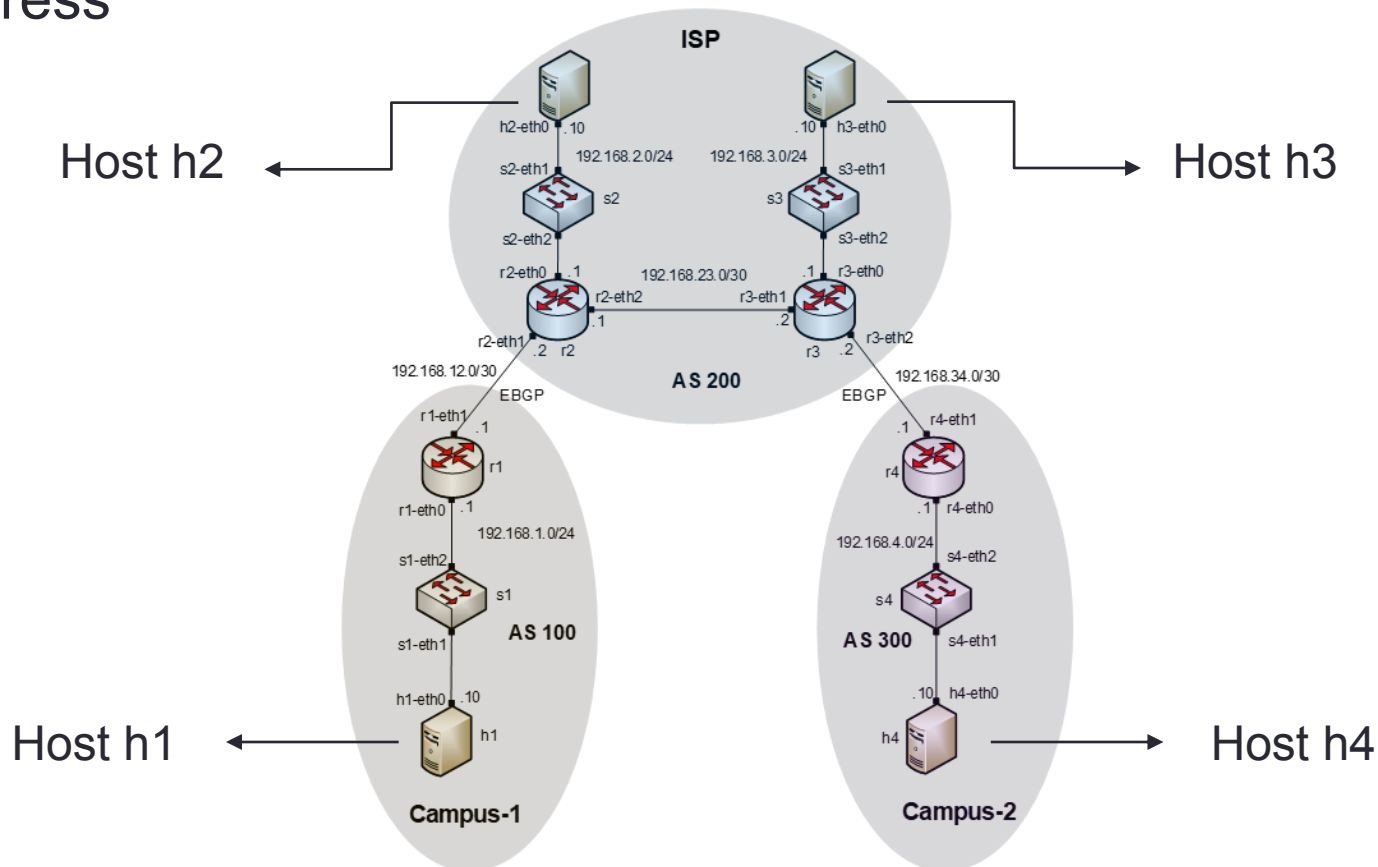


```

Host: h1
root@frr-pc:~# ifconfig lo 192.168.4.10
root@frr-pc:~#
  
```

Lab 12 Configuration

- Host h1 pings hosts h2 and h3 using the spoofed source IP address



```

Host: h1
root@frr-pc:~# fping --src 192.168.4.10 192.168.2.10 192.168.3.10
192.168.2.10 is unreachable
192.168.3.10 is unreachable
root@frr-pc:~#
  
```

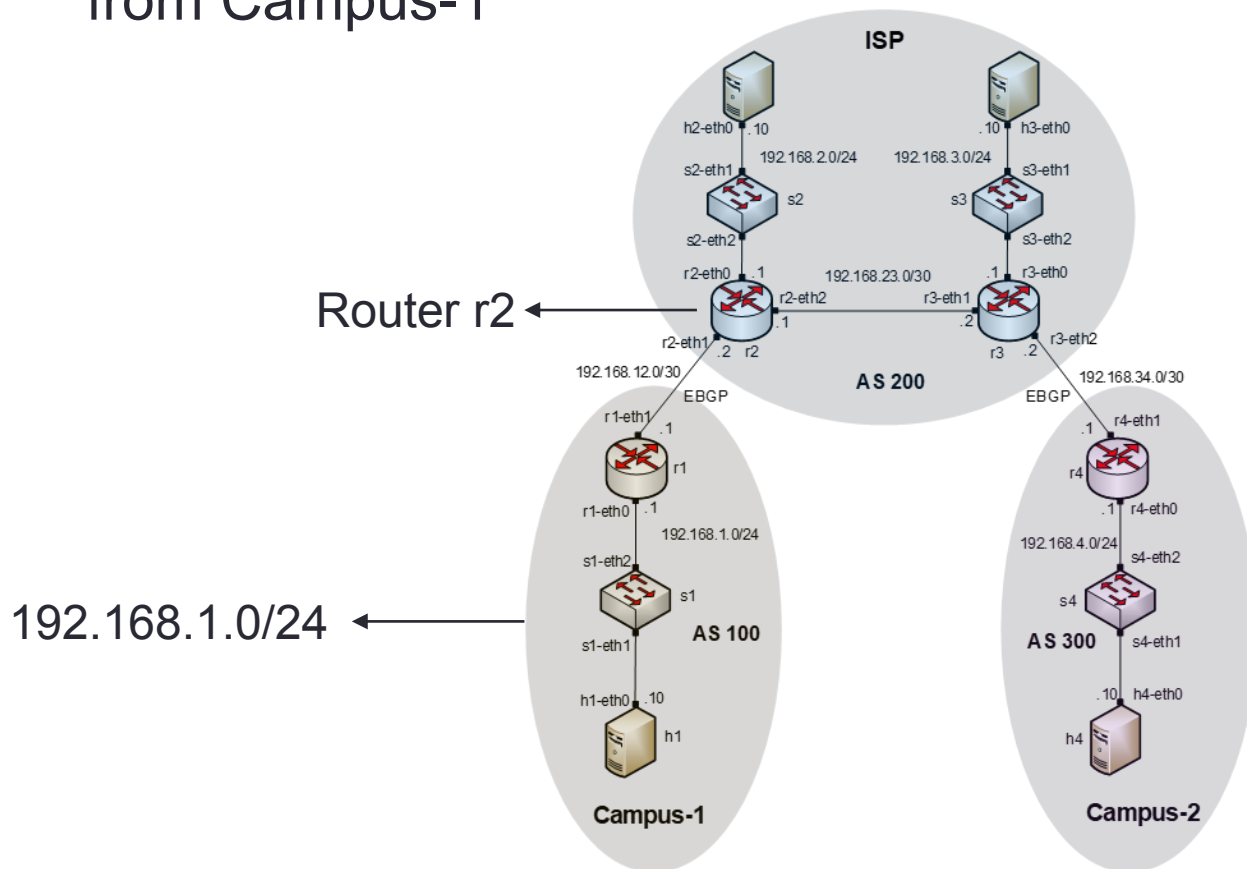
Lab 12 Configuration

- Host h4 receives a reply messages from 192.168.2.10 and 192.168.3.10

```
"Host: h4"
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on h4-eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
13:31:12.049239 IP6 fe80::b8f9:daff:fe95:76d0.mdns > ff02::fb.mdns: 0 [2q] PTR (
QM)? _ipps._tcp.local. PTR (QM)? _ipp._tcp.local. (45)
13:32:03.804279 IP 192.168.2.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 0,
length 64
13:32:03.814074 IP 192.168.3.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 1,
length 64
13:32:04.304134 IP 192.168.2.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 2,
length 64
13:32:04.314484 IP 192.168.3.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 3,
length 64
13:32:05.055159 IP 192.168.2.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 4,
length 64
13:32:05.065320 IP 192.168.3.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 5,
length 64
13:32:06.181389 IP 192.168.2.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 6,
length 64
13:32:06.191543 IP 192.168.3.10 > 192.168.4.10: ICMP echo reply, id 3837, seq 7,
length 64
13:32:08.940505 ARP, Request who-has 192.168.4.10 tell 192.168.4.1, length 28
13:32:08.940523 ARP, Reply 192.168.4.10 is-at 16:c3:6e:0f:a9:49 (oui Unknown), l
ength 28
```


Lab 12 Configuration

- Apply a filter on router r2 to accept IP source 192.168.1.0/24 from Campus-1

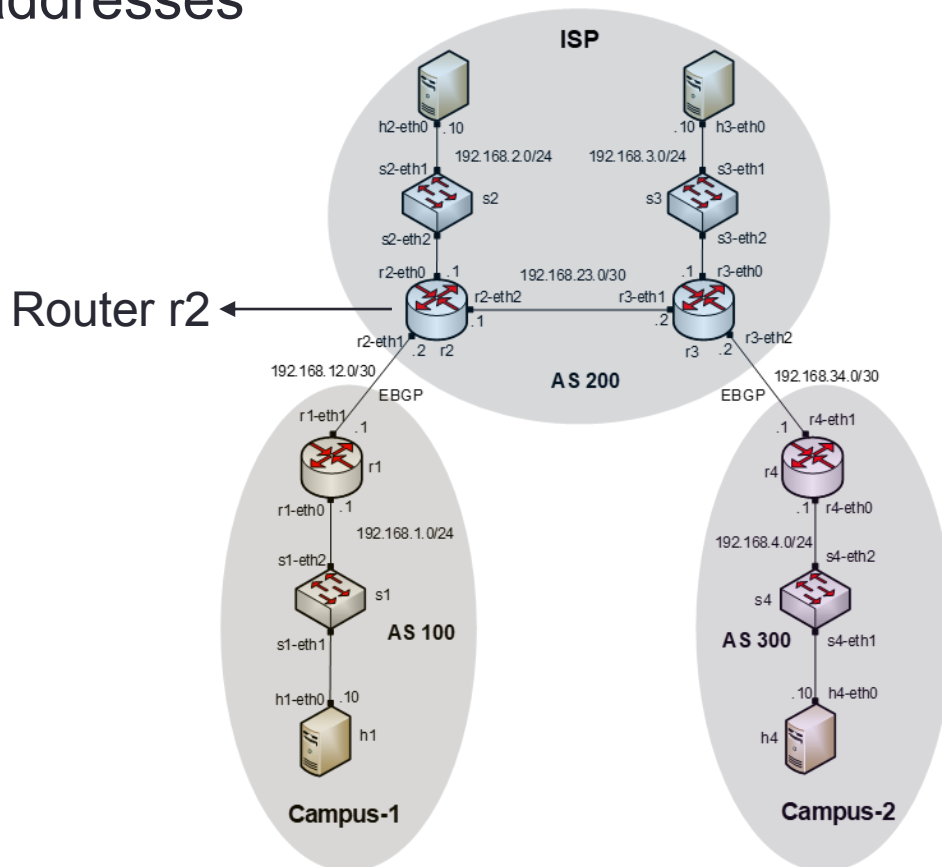


```

Host: r2
frr-pc# exit
root@frr-pc:/etc/routers/r2# iptables -A FORWARD -s 192.168.1.0/24 -i r2-eth1 -j ACCEPT
root@frr-pc:/etc/routers/r2#
  
```

Lab 12 Configuration

- Apply another route filter on router r2 to reject all other IP source addresses

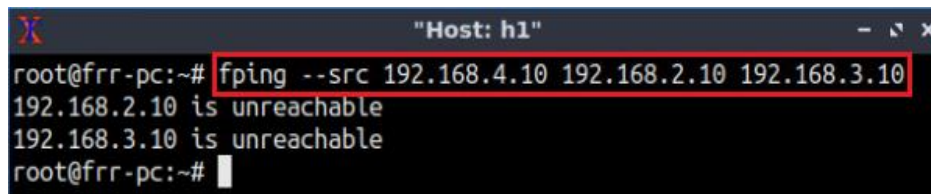


```

X                                     "Host: r2"
frr-pc# exit
root@frr-pc:/etc/routers/r2# iptables -A FORWARD -s 192.168.1.0/24 -i r2-eth1 -j ACCEPT
root@frr-pc:/etc/routers/r2# iptables -A FORWARD -s 0/0 -i r2-eth1 -j DROP
root@frr-pc:/etc/routers/r2#
  
```

Lab 12 Configuration

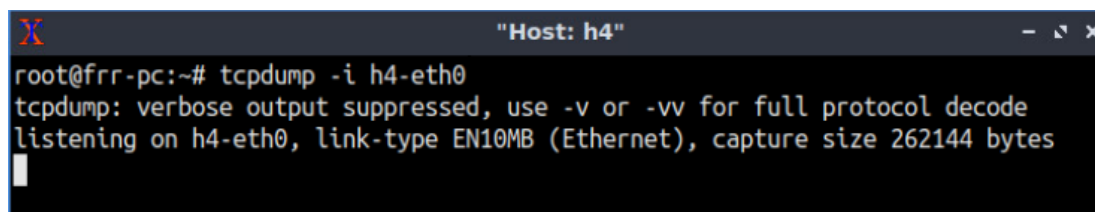
- Perform DoS attack by host h1 on host h4



```
root@frr-pc:~# fping --src 192.168.4.10 192.168.2.10 192.168.3.10
192.168.2.10 is unreachable
192.168.3.10 is unreachable
root@frr-pc:~#
```

A terminal window titled "Host: h1" showing a DoS attack using the `fping` command. The command `fping --src 192.168.4.10 192.168.2.10 192.168.3.10` is entered, and the output shows that both 192.168.2.10 and 192.168.3.10 are unreachable.

- Capture the network traffic on host h4



```
root@frr-pc:~# tcpdump -i h4-eth0
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on h4-eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

A terminal window titled "Host: h4" showing the configuration of `tcpdump` to capture network traffic on the `h4-eth0` interface. The output shows that the capture is listening on `h4-eth0` with a link-type of `EN10MB (Ethernet)` and a capture size of 262144 bytes.